

Migrating a JavaScript Application to TypeScript



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Gang of Four Patterns

Erich Gamma,
Richard Helm,
Ralph Johnson,
John Vlissides

Battlefield tactics

*Derive the tactic
from circumstances*



The Patterns

Observer

Pub-Sub





Our Client JavaScript App



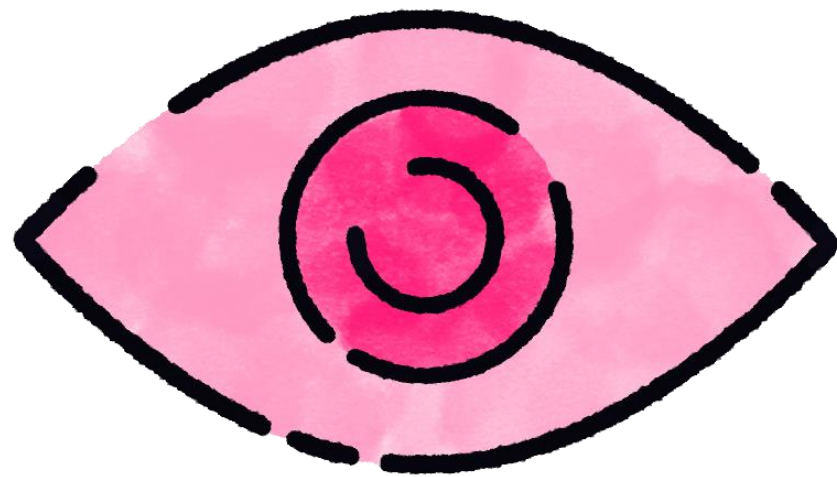
The Three Observers

Things we do with the data

Currently, they're
all tangled up together



The Base Observer



The base observer is very simple

`receiveNotification`

It will receive a `dataPacket` (at first)

Three observers:

- `ChartObserver`
- `TableObserver`
- `ConsoleObserver`

Adding a new one later on is very easy

We are *refactoring*

Moving from one working system to another



The background features a light gray abstract pattern with various shapes, including circles, squares, and wavy lines, some containing smaller patterns like dots or internal wavy lines.

Demo: Migrating Our Client JS to TypeScript



Demo: Migrating Our Client JS to TypeScript, Part II



Module Resolution Problems

Wow – this is really simple!

**We can struggle with this,
or just use webpack**



Two Primary Problems

**The name resolution strategies
for JavaScript and NodeJs
are in conflict**

***node_modules* doesn't work for
client-side**

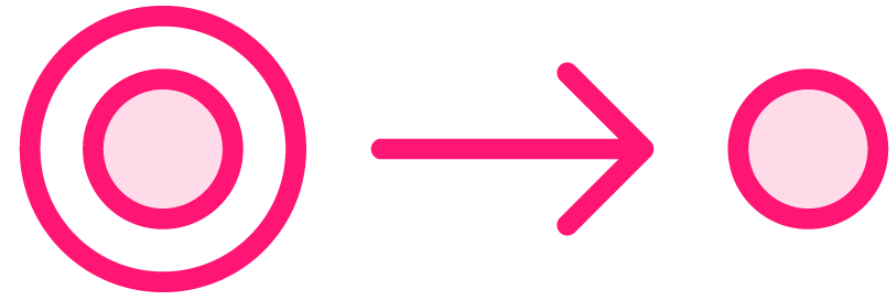


The background features a light gray abstract pattern with various shapes, including circles, squares, and wavy lines, some containing smaller patterns like dots or internal wavy lines.

Demo: Migrating Our Server JS to TypeScript



I Have Not Yet Begun to Refactor



I would continue with ChartObserver and make it more fluent

And more testable

Issue a known set of packets and check the svg against a known set

Could we have just done all of this in JavaScript? Yes...

But TypeScript represents all these ideas more clearly and directly



Thank you very much for watching!

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